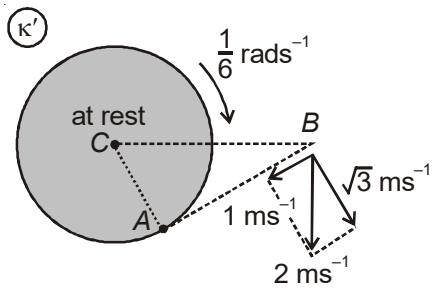
Physics

Answer Key

1. 

$2\ \text{ms}^{-1}$ in a direction perpendicular to the line joining him to the centre of the carousel.

2. Joe must paddle in a direction that makes an angle $\frac{\pi}{2}+\theta$ with the drift velocity of the water, where

$$\theta = \sin^{-1}\left(\frac{u}{v+u}\right). \qquad \ldots(1)$$

$v \geq u \sin\theta$. $\qquad \ldots(2)$

In the alternative situation ($v < u \sin\theta$),

$$\theta = \sin^{-1}\left(\frac{v}{u}\right). \qquad \ldots(3)$$

By equating the values of θ given by equations (1) and (2), we can obtain the critical value for $\frac{u}{v}$ that separates the two strategies:

$$\frac{u}{v} = \frac{\sqrt{5}+1}{2} \approx 1.618,$$

3. The minimal time for his turn is $t = \frac{|v_2 - v_1|}{\mu g} = \frac{\sqrt{2}v}{\mu g} \approx 7.2\ \text{s}$.

The boy's acceleration vector is constant throughout, with magnitude μg, and directed to the southeast (it makes an angle of 45° with the x-axis), and so his trajectory during the turn (in the same way as for projectiles) is a parabola, as shown in Fig. 2.

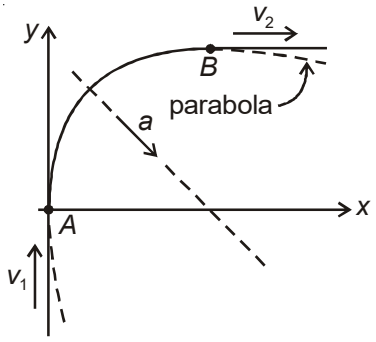


Fig. 2

4. *PB* is the arc that is traversed in a shorter time.

5. The smaller ball is lifted from the table-top immediately after the release.

6. $v_0 \approx \sqrt{\ell g}\ \theta_0 = \omega d$,

where $\omega = \sqrt{\frac{g}{\ell}}$ is the angular frequency of the pendulum; the other ball is, of course, at rest. The law of conservation of linear momentum then shows that, when the two balls are moving in phase, their maximal speed will be $\frac{v_0}{2}$, and so the (common) amplitude of their swings will be $\frac{d}{2}$.

7. $L_{\min} = \sqrt{\frac{m}{M}}\, L$.

8. The cord becomes slack and the weight goes into free fall!

9. The line of action of the impulse, and hence that of the cue, must be directed through point *P* (see Fig. 1).

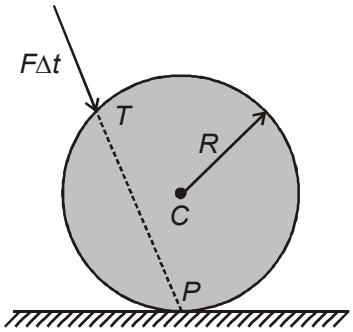


Fig. 1

The appropriate fixed point *Q* is shown in Fig. 2.

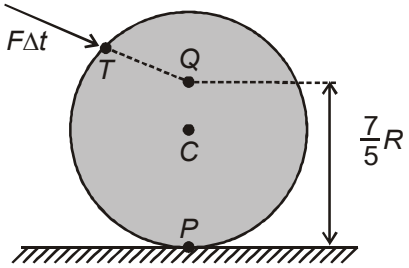


Fig. 2

The impulse is directed along the line *TQ* – the cue axis must follow the same line!

10. $W = Mg(s_2\mu_1 + s_1\mu_2)$.

11. $F = \frac{\lambda L^2 g}{8d}$.

12. (a) The centre of mass of the rolled rug is not exactly above the point of support – in the figure in the problem, it is slightly to the left of it – and so the rug's weight produces a torque about that point, which then causes the rug to unroll.

$$F = F(L) = \frac{3R}{2L}Mg.$$

13. $r_1 = \frac{L}{2} = 20$ cm.

$r_2 = L = 40$ cm.

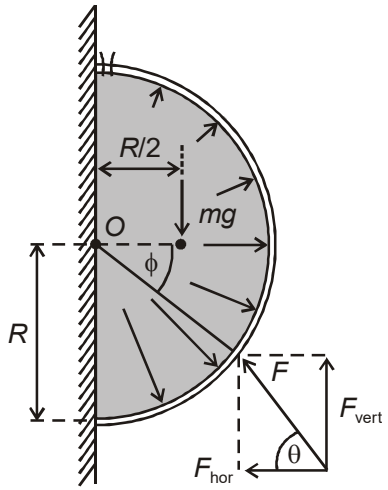
14. The magnitude of the required external force is therefore

$$F = \sqrt{F_{hor}^2 + F_{ver}^2} = Mg\sqrt{\frac{9}{4} + \left(1 + \frac{m}{M}\right)^2},$$

and its direction makes an angle

$$\theta = arc\tan\frac{F_{vert}}{F_{hor}} = arc\tan\frac{2(m+M)}{3M}$$

with the horizontal (see figure).



The required position of the force application point can be characterised by the angle ϕ shown in the figure. The remaining effective torques about O are then $\frac{mgR}{2}$ and $\frac{(3MGR\sin\phi)}{2}$, both clockwise, and $(m + M)gR\cos\phi$ anticlockwise. From their balance, it follows that

$$\left(1 + \frac{M}{m}\right)\cos\phi - \frac{1}{2} - \frac{3M}{2m}\sin\phi = 0.$$

This equation is quadratic in either $\sin\phi$ or $\cos\phi$, and can be solved if the ratio of the masses is known.

15. The water level must drop when the blade sinks.

16. $\rho = \frac{Rr}{R - r}$.

$$a = R\sin\theta = \frac{\sqrt{3}}{2}\frac{Rr}{\sqrt{R^2 + r^2 - Rr}}.$$

17. Suppose that we construct a container with two adjacent but separate compartments; all the outer walls are to be insulating, but the common connecting one is to be heat-conducting. One compartment is for tap water and the other for the distilled water, the only interaction between them being thermal conduction through the intervening wall.

For our analysis, we assume that the specific heats of distilled and tap water are equal, and so the ultimate

equilibrium temperature for any particular stage is the mass-weighted arithmetic mean of the temperatures of the components when the stage begins. For example, if distilled water of mass M and temperature T_0 (measured in degrees Celsius) and tap water of mass m and temperature 100°C are poured into the compartments, then the common equilibrium temperature will be

$$T_1 = \frac{MT_0 + m \times 100°C}{M + m}$$

For $M = m = 1$ kg, the common temperature would be 50°C, which is not good enough!

By dividing the hot tap water into a number, say six, of smaller equal portions, we can do better. If all the cold distilled water is put into one of the compartments of the heat exchanger, and $\frac{1}{6}$ kg of tap water at 100°C is placed into the other, then, after equalisation, the common temperature will, from the above formula, be

$$T_1 = \frac{1 \times 0 + \frac{1}{6} \times 100}{\frac{7}{6}} = 14.3°\text{C}.$$

We now pour away the cooled tap water, and refill its compartment with a further $\frac{1}{6}$ kg of water at 100°C. After the second heat exchange, the common temperature will be

$$T_2 = \frac{1 \times 14.3 + \frac{1}{6} \times 100}{\frac{7}{6}} = 26.5°\text{C}.$$

Continuing in this way, after the sixth and final heat exchange, the temperature of the distilled water will be $T_6 = 60.3$°C – and the task set in the problem has been accomplished!

18. $W = \frac{3}{2} nR \int_{T_0}^{2T_0} \left(1 - \frac{T}{2T_0}\right) dT = \frac{3}{8} nRT_0.$

19. $T_0 = \frac{T_1 + T_2 + T_3}{3}.$

20. $Q = 313$ kJ

21. So the pressure of the steam in the test tube in the final state is still 101 kPa!

22. $R = \frac{1}{\varepsilon}.$

23. The number of interference fringes observable on the screen is

$$N \approx \frac{D}{\Delta} = \frac{\delta^2}{\lambda} \frac{H + p}{H(p - f) - pf}.$$

Substituting the given data into the formula, we get the result $N \approx 46.7$, and so about 47 interference fringes are visible on the screen.

24. $2\pi k_e \lambda^2.$

25. $F^{(b)}$ $\frac{1}{8\pi\varepsilon_0} \frac{Qq}{R^2}.$

26. $-4\pi\varepsilon_0 V_0 \frac{rR}{\sqrt{r^2 + R^2}}.$

27. $B_{\parallel} = \dfrac{\mu_0 I R^2}{2L^3}$.

$B_{\perp} = \dfrac{\mu_0 I R^2}{4L^3}$.

28. $B(r) = \mu_0 \dfrac{I}{2\pi r}$.

29. $R = \dfrac{1}{\pi}\sqrt{\dfrac{\mu_0}{\varepsilon_0}} \ln 100 = 553\ \Omega$.

30. $I_A = \dfrac{I_0 + \sqrt{I_0^2 + 4I_1^2}}{2}$.

31. $L_3 = 3(L_2 - L_1)$.

32. The pearl moves continuously away from P and continues to do so at the end of the process.

❑ ❑ ❑

Chemistry

Answer Key

1. Answer (D)
2. Answer (B)
3. Answer (C)
4. Answer (B)
5. Answer (C)
6. Answer (D)
7. Answer (C)
8. Answer (A)
9. Answer (B)
10. Answer (D)
11. Answer (C)
12. Answer (A, C)
13. Answer (A, B, C)
14. Answer (A, B, C)
15. Answer (B, C)
16. Answer (B)
17. Answer (B)
18. Answer (D)
19. Answer (B)
20. Answer (C)
21. Answer (C)
22. Answer (A, B, C)
23. Answer (A, B, C)
24. Answer (A, C)
25. Answer (A, C, D)
26. Answer A $\rightarrow$ (P); B $\rightarrow$ (S); C $\rightarrow$ (R); D $\rightarrow$ (Q, T)
27. Answer (16)
28. Answer (1)
29. Answer (B)
30. Answer (C)
31. Answer (B)
32. Answer (B)
33. Answer (D)
34. Answer (B)
35. Answer (C)
36. Answer (C)
37. Answer (B)
38. Answer (B)
39. Answer (D)
40. Answer (A)
41. Answer (D)
42. Answer (D)
43. Answer (D)
44. Answer (B)
45. Answer (D)
46. Answer (B, D)
47. Answer (9)
48. Answer (28)
49. Answer (16)
50. Answer (03)
51. Answer (D)
52. Answer (B)
53. Answer (B)
54. Answer (C)
55. Answer (A)
56. Answer (C)
57. Answer (B)
58. Answer (D)
59. Answer (A, B, D)
60. Answer (A, B, D)

61. Answer (A, B, D)
62. Answer (B, D)
63. Answer (B, C)
64. Answer (41)
65. Answer (1)
66. Answer (B)
67. Answer (B)
68. Answer (B)
69. Answer (C)
70. Answer (B)
71. Answer (B)
72. Answer (B)
73. Answer (C)
74. Answer (D)
75. Answer (B)
76. Answer (D)
77. Answer (A $\rightarrow$ P, Q, R); (B $\rightarrow$ P, Q); (C $\rightarrow$ S, T); (D $\rightarrow$ P, S)
78. Answer (B)
79. Answer (C)
80. Answer (B)
81. Answer (B)
82. Answer (B)
83. Answer (A)
84. Answer (D)
85. Answer (C)
86. Answer (A)
87. Answer (B)
88. Answer (C)
89. Answer (D)
90. Answer (D)
91. Answer (C)
92. Answer (A)
93. Answer (B)
94. Answer (C)
95. Answer (B)
96. Answer (A)
97. Answer (A)
98. Answer (C)
99. Answer (C)
100. Answer (D)
101. Answer (A)
102. Answer (A, B, C)
103. Answer (A, B, C)
104. Answer (A $\rightarrow$ Q); (B $\rightarrow$ P); (C $\rightarrow$ S); (D $\rightarrow$ R)
105. Answer (1)
106. Answer (9)
107. Answer (6)
108. Answer (C)
109. Answer (C)
110. Answer (B)
111. Answer (C)
112. Answer (D)
113. Answer (C)
114. Answer (A)
115. Answer (D)
116. Answer (C)
117. Answer (C)
118. Answer (D)
119. Answer (C)
120. Answer (B)
121. Answer (A, B, C, D)
122. Answer (C)
123. Answer (A, D)
124. Answer (0)
125. Answer (C)
126. Answer (A)
127. Answer (B)
128. Answer (A)
129. Answer (D)
130. Answer (D)
131. Answer (D)
132. Answer (A)
133. Answer (A)
134. Answer (C)
135. Answer (B)
136. Answer (C)
137. Answer (B, C)
138. Answer (3)

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MATHEMATICS

Answer Key

1.	(3)	2.	(2)	3.	(3)	4.	(2)	5.	(3)	6.	(2)
7.	(1)	8.	(3)	9.	(1, 3)	10.	(1, 2, 3)	11.	(1, 4)	12.	(1)
13.	(3)										
14.	A(R); B(S); C(P); D(Q)										
15.	(26)	16.	(6)	17.	(6)	18.	(4)	19.	(4)	20.	(2)
21.	(4)	22.	(4)	23.	(4)	24.	(1)	25.	(1, 2, 3)	26.	(1, 3, 4)
27.	(1, 2)	28.	(0)	29.	(1)	30.	(1)	31.	(3)	32.	(3)
33.	(2)	34.	(3)	35.	(2)	36.	(2)	37.	(3)	38.	(4)
39.	(3, 4)	40.	(1, 4)	41.	(1, 3)	42.	(2, 3)	43.	(1, 2, 3, 4)	44.	(2)
45.	(2)										
46.	A(P, Q, R, S, T); B(P, T); C(Q, S); D(P, R, T)										
47.	(17)	48.	(3)	49.	(3)	50.	(4)	51.	(4)	52.	(1)
53.	(2)	54.	(1, 3, 4)	55.	(3, 4)	56.	(1, 2, 4)	57.	(1, 2, 3, 4)	58.	(4)
59.	(3)	60.	(2)	61.	(2)	62.	(1)	63.	(4)	64.	(4)
65.	(1)	66.	(4)	67.	(4)	68.	(2)	69.	(4)	70.	(4)
71.	(4)	72.	(2)	73.	(4)	74.	(2, 3)	75.	(1, 2)	76.	(3)
77.	(4)										
78.	A(S); B(P); C(P); D(S)										
79.	(7)	80.	(5)	81.	(2)	82.	(5)	83.	(2)	84.	(2)
85.	(2)	86.	(1)	87.	(4)	88.	(2)				
89.	A(R); B(Q); C(P); D(T)										
90.	(6)	91.	(4)	92.	(2)	93.	(3)	94.	(2)	95.	(3)
96.	(2)	97.	(3)	98.	(4)	99.	(3)	100.	(1)	101.	(4)
102.	(1)	103.	(3)	104.	(3)	105.	(2)	106.	(2)	107.	(2)
108.	(4)	109.	(3)	110.	(1, 3)	111.	(1, 2, 3)	112.	(1, 3, 4)	113.	(1, 3)
114.	(2)	115.	(4)								
116.	A(Q); B(P); C(S); D(R)										
117.	(5)	118.	(8)	119.	(5)	120.	(1)	121.	(3)	122.	(2)
123.	(4)	124.	(2)	125.	(4)	126.	(4)	127.	(1, 2, 3, 4)	128.	(1, 2)
129.	(2, 3)	130.	(4)	131.	(2)						
132.	A(S); B(P); C(Q); D(R)										

133. (3) 134. (8) 135. (2) 136. (2) 137. (3) 138. (1, 2, 4)
139. (2) 140. (4) 141. (2) 142. (1) 143. (3) 144. (2, 3, 4)
145. (4) 146. (3) 147. (1)
148. A(S); B(T); C(S); D(Q, R, S, T)
149. (4) 150. (0) 151. (2) 152. (1) 153. (1, 2)
154. A(S); B(R); C(P); D(Q)
155. (3) 156. (4) 157. (2) 158. (3) 159. (4) 160. (1, 4)
161. A(Q); B(R); C(P); D(T)
162. (5) 163. (7) 164. (7) 165. (3) 166. (1) 167. (1)
168. (2, 4)
169. A(Q); B(R); C(S); D(P)
170. (1) 171. (2) 172. (4) 173. (2) 174. (3) 175. (4)
176. (1, 3, 4) 177. (2) 178. (7) 179. (3) 180. (3) 181. (4)
182. (2) 183. (4) 184. (3) 185. (1) 186. (3)
187. A(R); B(S); C(P); D(P)
188. (93) 189. (7) 190. (3) 191. (2) 192. (2) 193. (3)
194. (8) 195. (2) 196. (3) 197. (3) 198. (3) 199. (4)
200. (1) 201. (4)
202. A(P); B(Q); C(R); D(S)
203. (25) 204. (4) 205. (1) 206. (1) 207. (3) 208. (3)
209. (4) 210. (1) 211. (2) 212. (3) 213. (2) 214. (2)
215. (2) 216. (1) 217. (1) 218. (1, 2, 4) 219. (1, 4)
220. A(S); B(R); C(Q); D(P)
221. (2) 222. (9) 223. (1) 224. (1) 225. (2) 226. (2, 3, 4)
227. A(S); B(P, T); C(R); D(T)
228. (8) 229. (2) 230. (1) 231. (1) 232. (4) 233. (1)
234. (1) 235. (1) 236. (1, 3) 237. (1, 2) 238. (3) 239. (3)
240. (2) 241. (2) 242. (2) 243. (3) 244. (4) 245. (1)
246. (1) 247. (1, 2, 4) 248. (4) 249. (1) 250. (3) 251. (1)
252. (2) 253. (4) 254. (2, 3, 4) 255. (1, 4) 256. (1, 4) 257. (1, 2, 3, 4)
258. (4) 259. (3) 260. (4) 261. (2) 262. (2) 263. (8)

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